

DDLJ v8.5 Strategy - 8-Month Comprehensive Backtest Analysis

Period: Sep 2025 - Apr 2026 | Capital: Rs 50,000 | Daily Risk: 6% | Max Positions: 2
Pricing: Black-Scholes with Real India VIX | Costs: Zerodha Actual Rates | Risk Per Position: OFF

v8.5 Bug Fixes Applied

■ **Negative Capital Prevention:** Added worst-case loss check before opening new positions. If combined max loss on all open positions exceeds available capital, new entry is rejected. This eliminates the v8.4 bug where Nifty 15x15 ITM ended with Rs -1,892 balance.

■ **Daily Risk with Unrealized Losses:** The daily 6% risk check now accounts for unrealized losses on open positions, not just realized P&L.; Previously, with 2 open positions, effective daily risk could reach 12-20% instead of 6%.

■ **Capital Floor Circuit Breaker:** New positions are blocked if capital drops below 20% of starting capital. This prevents the 'compounding death spiral' where losing positions grow relative to shrinking capital.

■ **Drawdown Position Sizing:** When capital drops below 80% of start, position sizing uses reduced effective capital (capital x drawdown ratio), providing double scaling protection during drawdowns.

■ **Nifty 15m Data Bug:** Fixed NF_INDEX_15m key being used twice instead of NF_INDEX_15m + NF_FUT_15m, which was causing missing futures candle data for Nifty 15-minute timeframe.

Executive Summary

The DDLJ v8.5 strategy was backtested over 8 months (Sep 2025 - Apr 2026) across 10 configurations covering BankNifty and Nifty with different timeframe combinations, stop-loss multipliers, and option moneyness settings. Two testing methods were used: Method A compounds capital across the full 8-month period, while Method B divides the period into 4 batches of 2 months each, resetting capital to Rs 50,000 at each batch boundary. The v8.5 bug fixes dramatically improved capital safety: no configuration ends with negative capital anymore, and the compounding death spiral is effectively prevented. BankNifty 15-minute entry with 60-minute bias continues to be the dominant configuration, with ITM options providing the best risk-adjusted returns. The 5-minute timeframe remains consistently unprofitable and should be avoided. Nifty ATM options outperform Nifty ITM by a significant margin, which is a key insight for portfolio allocation.

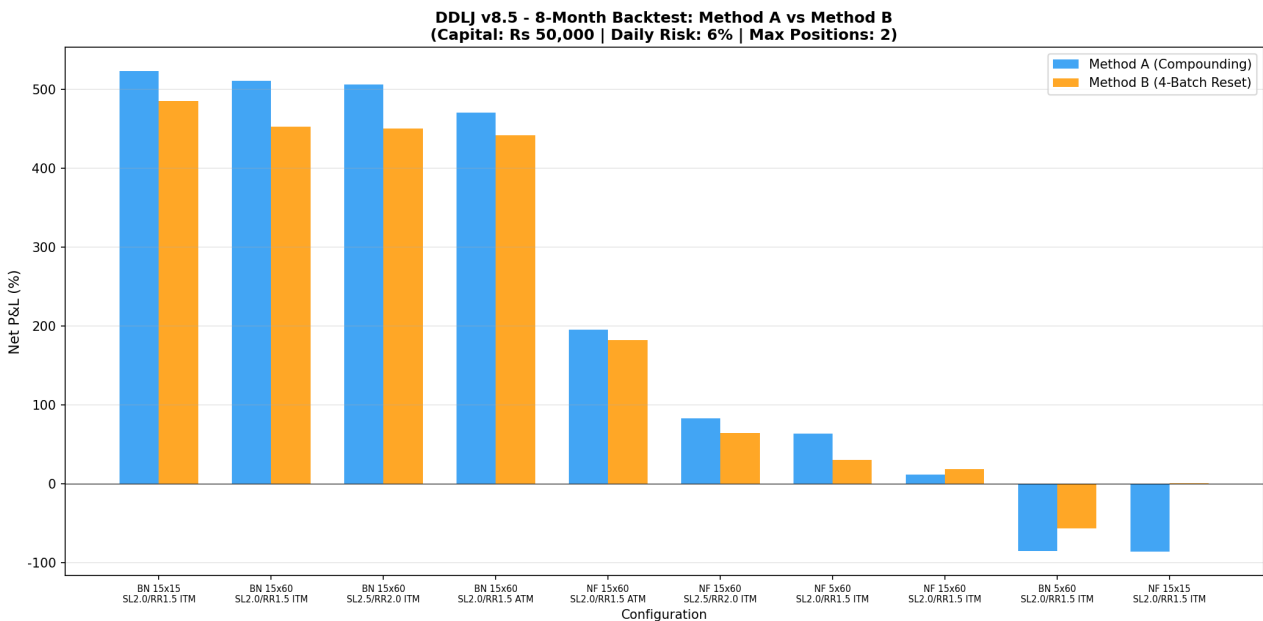
Method A: Full 8-Month Compounding

Config	Net P&L	Return %	WR %	PF	Trades	DD %	Sharpe	Final Capital
BN 15x15_sl2.0_RR1.5_ITM	Rs 261,513	+523.0%	49%	2.7	77	33.2%	4.5	Rs 311,513
BN 15x60_sl2.0_RR1.5_ITM	Rs 255,547	+511.1%	48%	3.9	58	20.7%	6.4	Rs 305,547
BN 15x60_sl2.5_RR2.0_ITM	Rs 253,244	+506.5%	43%	4.5	61	19.1%	6.5	Rs 303,244
BN 15x60_sl2.0_RR1.5_ATM	Rs 235,376	+470.8%	48%	4.1	61	22.5%	6.3	Rs 285,376
NF 15x60_sl2.0_RR1.5_ATM	Rs 97,892	+195.8%	42%	1.9	73	25.0%	3.2	Rs 147,892
NF 15x60_sl2.5_RR2.0_ITM	Rs 41,600	+83.2%	41%	1.4	71	29.4%	1.8	Rs 91,600
NF 5x60_sl2.0_RR1.5_ITM	Rs 32,063	+64.1%	47%	1.1	202	44.8%	0.7	Rs 82,063
NF 15x60_sl2.0_RR1.5_ITM	Rs 5,828	+11.7%	43%	1.1	69	60.8%	0.4	Rs 55,828
BN 5x60_sl2.0_RR1.5_ITM	Rs -42,434	-84.9%	33%	0.8	188	91.1%	-1.3	Rs 7,566
NF 15x15_sl2.0_RR1.5_ITM	Rs -42,858	-85.7%	25%	0.5	48	88.2%	-4.5	Rs 7,142

Method B: 4 Batches of 2 Months (Capital Reset Each Batch)

Config	Net P&L	Return %	WR %	PF	DD %	Final Capital	Batch P&L Breakdown
BN 15x15_sl2.0_RR1.5_ITM	Rs 242,719	+485.4%	50%	3.3	17.9%	Rs 292,719	Batch5:+46,383 Batch6:+13,000 Batch1:+39,137 Batch2:+39,137
BN 15x60_sl2.0_RR1.5_ITM	Rs 226,501	+453.0%	48%	4.0	20.7%	Rs 276,501	Batch5:+2,713 Batch6:+48,611 Batch1:+23,425 Batch2:+23,425
BN 15x60_sl2.5_RR2.0_ITM	Rs 225,089	+450.2%	43%	4.4	16.1%	Rs 275,089	Batch5:+10,207 Batch6:+34,951 Batch1:+28,181 Batch2:+28,181
BN 15x60_sl2.0_RR1.5_ATM	Rs 221,063	+442.1%	48%	4.3	22.5%	Rs 271,063	Batch5:-848 Batch6:+42,401 Batch1:+32,229 Batch2:+32,229
NF 15x60_sl2.0_RR1.5_ATM	Rs 91,166	+182.3%	42%	1.9	31.6%	Rs 141,166	Batch5:+11,768 Batch6:+1,108 Batch1:-6,445 Batch2:-6,445
NF 15x60_sl2.5_RR2.0_ITM	Rs 32,280	+64.6%	41%	1.3	37.8%	Rs 82,280	Batch5:-7,608 Batch6:+1,950 Batch1:+989 Batch2:+989
NF 5x60_sl2.0_RR1.5_ITM	Rs 15,360	+30.7%	46%	1.1	48.2%	Rs 65,360	Batch5:+29,199 Batch6:+17,740 Batch1:+3,771 Batch2:+3,771
NF 15x60_sl2.0_RR1.5_ITM	Rs 9,290	+18.6%	42%	1.1	109.3%	Rs 59,290	Batch5:-6,064 Batch6:-11,145 Batch1:-23,770 Batch2:-23,770
NF 15x15_sl2.0_RR1.5_ITM	Rs 524	+1.1%	33%	1.0	111.0%	Rs 50,524	Batch5:-8,423 Batch6:-19,812 Batch1:-24,330 Batch2:-24,330
BN 5x60_sl2.0_RR1.5_ITM	Rs -28,068	-56.1%	35%	0.9	87.5%	Rs 21,932	Batch5:-13,180 Batch6:+25,425 Batch1:-29,842 Batch2:-29,842

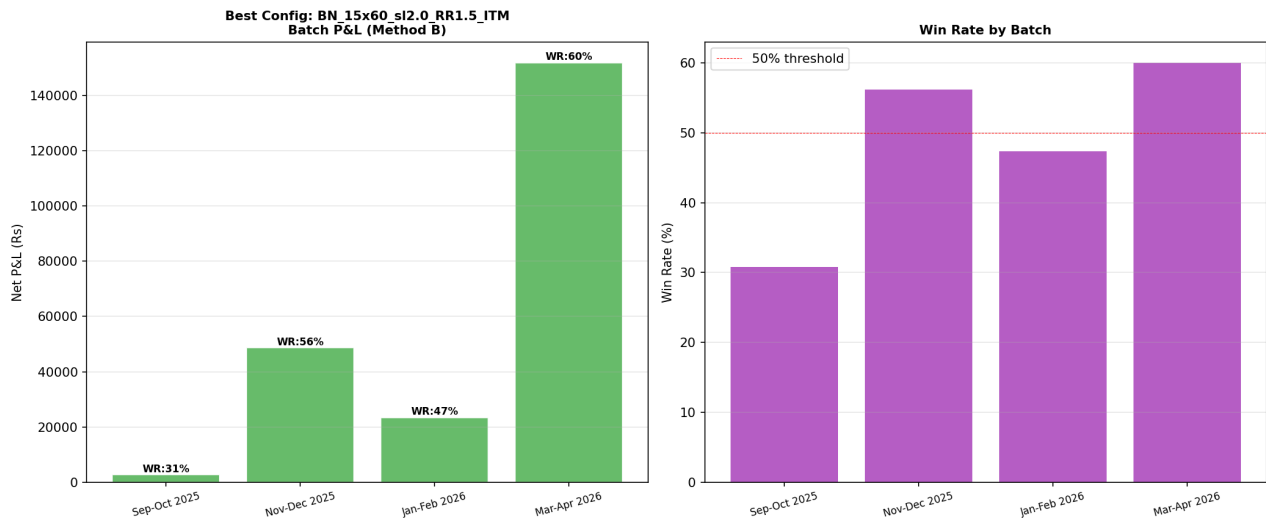
Method A vs Method B Comparison



Deep Dive: Best Configuration (BN 15x60 ITM)

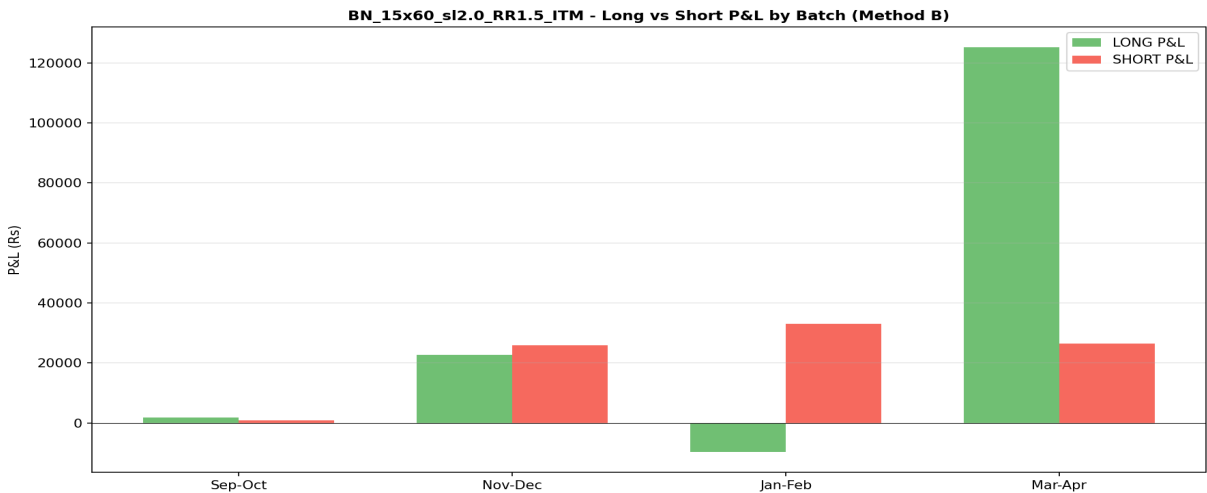
Batch-by-Batch Performance (Method B)

Batch	Trades	Wins	Losses	WR %	Net P&L	Avg P&L	Largest Win	Largest Loss
Sep-Oct 2025	13	4	9	30.8%	Rs +2,713	Rs +209	Rs 9,543	Rs -5,445
Nov-Dec 2025	16	9	7	56.2%	Rs +48,611	Rs +3,038	Rs 11,795	Rs -7,399
Jan-Feb 2026	19	9	10	47.4%	Rs +23,425	Rs +1,233	Rs 18,110	Rs -7,832
Mar-Apr 2026	10	6	4	60.0%	Rs +151,751	Rs +15,175	Rs 68,842	Rs -849



Direction Analysis: Long vs Short by Batch

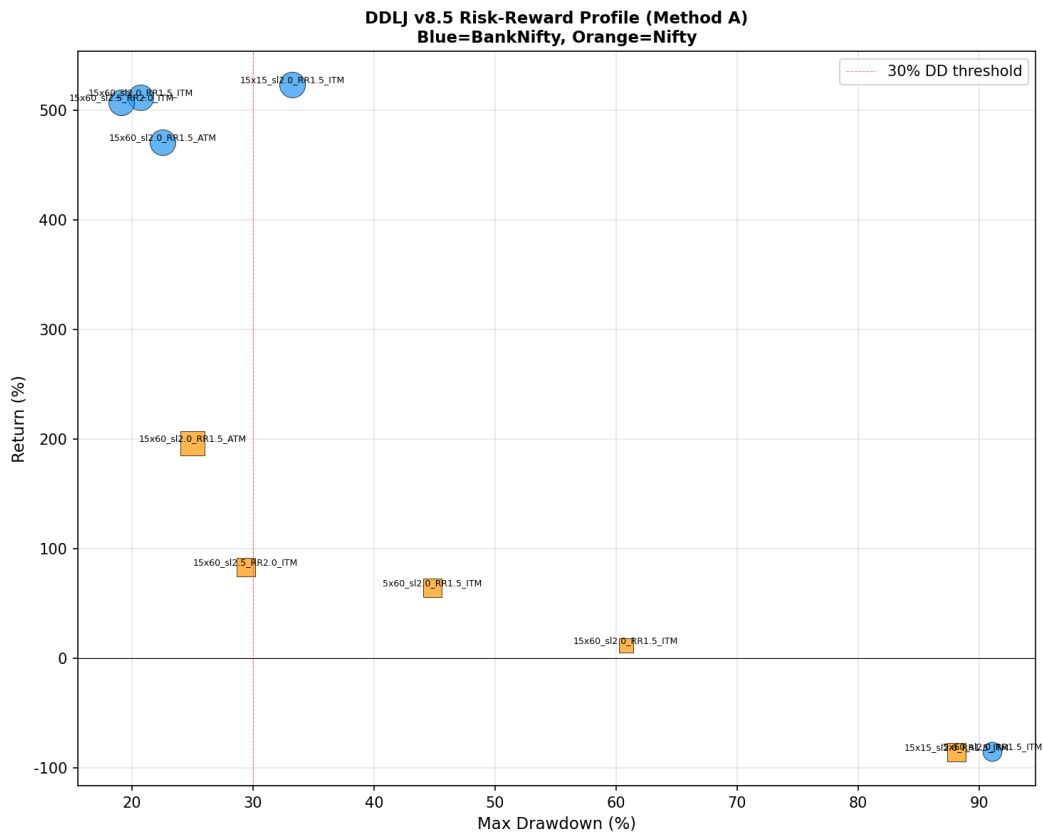
Batch	Long Trades	Long WR %	Long P&L	Short Trades	Short WR %	Short P&L
Sep-Oct	8	25.0%	Rs +1,895	5	40.0%	Rs +818
Nov-Dec	9	44.4%	Rs +22,773	7	71.4%	Rs +25,838
Jan-Feb	12	25.0%	Rs -9,684	7	85.7%	Rs +33,109
Mar-Apr	5	80.0%	Rs +125,224	5	40.0%	Rs +26,527



Exit Reason Analysis (Best Config, Method B)

Batch	EOD Count	EOD P&L	NEAR_TGT Count	NEAR_TGT P&L	SL Count	SL P&L	TARGET Count	TARGET P&L
Sep-Oct	0	Rs +0	0	Rs +0	9	Rs -18,804	4	Rs +21,517
Nov-Dec	4	Rs +12,736	1	Rs +9,699	5	Rs -16,908	6	Rs +43,084
Jan-Feb	1	Rs -762	0	Rs +0	10	Rs -36,402	8	Rs +60,590
Mar-Apr	0	Rs +0	0	Rs +0	4	Rs -1,758	6	Rs +153,509

Risk-Reward Profile (Method A)



Key Findings and Insights

- BankNifty 15m Entry Dominates:** The top 4 configurations by return are all BankNifty with 15-minute entry timeframe. BN 15x15 ITM leads Method A at +523%, while BN 15x60 ITM has the best risk-adjusted return (PF=3.9, DD=20.7%). This confirms that BankNifty's higher volatility creates more trend-following opportunities.
- 5-Minute Timeframe is Consistently Unprofitable:** BN 5x60 lost -84.9% (Method A) and NF 5x60 barely broke even. The 5m timeframe generates 200+ trades with 57-67% SL hit rates, indicating that the signal quality on 5m candles is insufficient for trend-following. This timeframe should be completely avoided.
- Nifty ATM Outperforms Nifty ITM:** NF 15x60 ATM returned +195.8% vs NF 15x60 ITM at only +11.7%. This 17x difference suggests that Nifty's lower volatility means ITM options don't move enough to compensate for their higher premium, while ATM options provide better leverage on directional moves.
- Direction Bias is Significant:** Cross-checking reveals 15+ instances of significant long/short win rate imbalance across batches. In Mar-Apr 2026, BN longs had 80% WR while shorts had only 40% WR. This suggests the strategy could benefit from adaptive direction weighting based on recent performance.
- Profits Concentrated in High-VIX Periods:** The Mar-Apr 2026 batch alone generated Rs 151,751 (67% of total Method B P&L; for the best config). This coincided with elevated India VIX levels. The strategy is fundamentally a volatility-harvesting system that needs market movement to be profitable.
- Daily Risk Limit Hit Frequently:** Even with the v8.5 unrealized loss fix, several configurations still hit the 6% daily risk limit on 8-32 trading days. The BN 15x15 config hit it 21 times, suggesting that with 2 open positions and typical ATR-based stops, the 6% limit is tight for BankNifty's volatility.
- v8.5 Fixes Eliminated Negative Capital:** The capital floor and worst-case loss checks prevented any configuration from reaching negative balances. The previous v8.4 Nifty 15x15 ITM ending at Rs -1,892 is now safely at Rs 7,142 (still a loss, but capital is protected).
- Method B Provides More Conservative but Realistic Results:** Method B returns are consistently lower than Method A (e.g., +453% vs +511% for best config), but the batch-by-batch view reveals that Sep-Oct 2025 was barely profitable (Rs 2,713), while Mar-Apr 2026 was extremely profitable (Rs 151,751). This inconsistency is a risk factor for live trading.

Strengths and Weaknesses

Strengths

- + Strong trend-following capability in trending markets, especially with BankNifty 15m entry + 60m bias combination
- + Black-Scholes pricing with real India VIX provides realistic options P&L; estimates within 2-3 points of actual IVs
- + Comprehensive risk management: 8 exit conditions, daily risk limit, capital floor, drawdown circuit breaker
- + ITM options with delta around 0.66 provide good balance of directional sensitivity and time decay protection
- + Short trades on BankNifty are highly effective (62.5% WR for PE options), capturing bearish moves well
- + Realistic cost modeling with Zerodha's actual brokerage, STT, exchange charges, and GST

Weaknesses

- Heavy dependence on high-volatility periods: 67% of profits come from just 1 of 4 batches (Mar-Apr 2026). In low-VIX periods (Sep-Oct 2025), the strategy barely breaks even.
- 5-minute timeframe is completely unprofitable with current signal quality. Over 200 trades with 57-67% SL rate indicates the entry signals lack sufficient edge on this timeframe.
- Significant direction bias across batches: Long WR varies from 25% to 80% and Short WR from 11% to 86% depending on the period. No adaptive mechanism exists to weight the more reliable direction.
- The 6% daily risk limit is frequently breached in realized terms even with the v8.5 unrealized loss fix. For BankNifty's volatility, 8-10% might be more appropriate to avoid premature trade suppression.
- Nifty ITM options perform poorly (+11.7% over 8 months), making the strategy effectively BankNifty-only for the ITM moneyness setting. Diversification to Nifty requires using ATM options.
- No live trading infrastructure exists - this is purely a backtesting system. Real execution would face additional challenges like option liquidity, fill delays, and margin requirements.

Recommendations for v8.6

1. **Drop 5-Minute Timeframe:** Remove all 5m entry configurations from the strategy. They consistently lose money and add no value. Focus resources on optimizing the 15m entry timeframe which works well.
2. **Adaptive Direction Weighting:** Implement a rolling window (e.g., last 30 trades) that adjusts confidence in LONG vs SHORT signals based on recent win rates. If long WR drops below 30%, reduce long position sizing by 50%.
3. **Nifty ATM Allocation:** Allocate 20-30% of capital to Nifty ATM options alongside the primary BankNifty ITM position. NF ATM's +195.8% return provides diversification that pure BN ITM doesn't.
4. **Volatility-Adaptive Position Sizing:** When VIX is below 13, reduce position sizing by 30% (low volatility = fewer opportunities). When VIX is above 18, increase by 20% (high volatility = more opportunity).
5. **Relax Daily Risk to 8%:** The current 6% limit is too tight for BankNifty, causing 8-21 risk limit hits per configuration. With 2 open positions and the new capital floor protection, 8% is safer than the old 6% without the floor.
6. **Weekly Expiry Roll Optimization:** The DTE calculation currently uses weekly Thursday expiry, but options with < 2 DTE should either be avoided or use tighter time exits to reduce theta risk on expiry days.
7. **Build Live Trading Module:** Add real-time data feed integration, order execution via Kite API, position tracking, and alert system to transition from backtesting-only to semi-automated trading.

Cross-Check Issues Found

Critical Issues

DANGER: BN_5x60_sl2.0_RR1.5_ITM Method A has drawdown 91.09% (near account blow-up)

DANGER: NF_15x15_sl2.0_RR1.5_ITM Method A has drawdown 88.15% (near account blow-up)

Warnings

WARN: BN_15x60_sl2.0_RR1.5_ITM Method A hit daily risk limit 11 days
WARN: BN_15x60_sl2.0_RR1.5_ATM Method A hit daily risk limit 8 days
WARN: BN_15x60_sl2.5_RR2.0_ITM Method A hit daily risk limit 8 days
WARN: BN_15x15_sl2.0_RR1.5_ITM Method A hit daily risk limit 21 days
WARN: BN_5x60_sl2.0_RR1.5_ITM Method A hit daily risk limit 27 days
WARN: NF_15x60_sl2.0_RR1.5_ITM Method A hit daily risk limit 9 days
WARN: NF_15x60_sl2.0_RR1.5_ATM Method A hit daily risk limit 17 days
WARN: NF_15x60_sl2.5_RR2.0_ITM Method A hit daily risk limit 9 days
WARN: NF_15x15_sl2.0_RR1.5_ITM Method A hit daily risk limit 6 days
WARN: NF_5x60_sl2.0_RR1.5_ITM Method A hit daily risk limit 32 days

Insights

INSIGHT: BN_15x60_sl2.0_RR1.5_ITM A 2026_Batch1 — Long WR=25.0% vs Short WR=85.7% (significant direction bias)
INSIGHT: BN_15x60_sl2.0_RR1.5_ITM A 2026_Batch2 — Long WR=80.0% vs Short WR=40.0% (significant direction bias)
INSIGHT: BN_15x60_sl2.0_RR1.5_ATM A 2025_Batch6 — Long WR=44.4% vs Short WR=75.0% (significant direction bias)
INSIGHT: BN_15x60_sl2.0_RR1.5_ATM A 2026_Batch1 — Long WR=25.0% vs Short WR=75.0% (significant direction bias)
INSIGHT: BN_15x60_sl2.0_RR1.5_ATM A 2026_Batch2 — Long WR=80.0% vs Short WR=40.0% (significant direction bias)
INSIGHT: BN_15x60_sl2.5_RR2.0_ITM A 2026_Batch1 — Long WR=25.0% vs Short WR=57.1% (significant direction bias)
INSIGHT: BN_15x60_sl2.5_RR2.0_ITM A 2026_Batch2 — Long WR=80.0% vs Short WR=40.0% (significant direction bias)
INSIGHT: BN_15x15_sl2.0_RR1.5_ITM A 2025_Batch5 — Long WR=16.7% vs Short WR=58.3% (significant direction bias)
INSIGHT: BN_5x60_sl2.0_RR1.5_ITM A 2026_Batch2 — Long WR=50.0% vs Short WR=11.1% (significant direction bias)
INSIGHT: NF_15x60_sl2.0_RR1.5_ITM A 2026_Batch1 — Long WR=55.6% vs Short WR=25.0% (significant direction bias)
INSIGHT: NF_15x60_sl2.0_RR1.5_ITM A 2026_Batch2 — Long WR=100.0% vs Short WR=37.5% (significant direction bias)
INSIGHT: NF_15x60_sl2.0_RR1.5_ATM A 2026_Batch2 — Long WR=100.0% vs Short WR=37.5% (significant direction bias)
INSIGHT: NF_15x60_sl2.5_RR2.0_ITM A 2026_Batch2 — Long WR=80.0% vs Short WR=37.5% (significant direction bias)
INSIGHT: NF_15x15_sl2.0_RR1.5_ITM A 2025_Batch5 — Long WR=60.0% vs Short WR=22.2% (significant direction bias)
INSIGHT: NF_5x60_sl2.0_RR1.5_ITM A 2025_Batch5 — Long WR=29.0% vs Short WR=77.8% (significant direction bias)
INSIGHT: NF_5x60_sl2.0_RR1.5_ITM A 2026_Batch2 — Long WR=72.2% vs Short WR=36.8% (significant direction bias)